MODEL RESEARCH

1. What is a Model in our Project?

For HR data, a Model is a mathematical engine that learns patterns from employee data to predict the next outcome.

ML Models: Learn from structured tables (Excel/SQL). Ex: Age, Salary, Department -> Will employee leave?

AI / LLM Models: Learn from unstructured text. Ex: Employee feedback -> Sentiment positive/negative?

Tokens: LLMs don't read words, they read tokens.

Transformer: Architecture from 2017 that helps model remember context across long HR documents.

2. How ML & AI Models are Taught

For ML Models (XGBoost, Random Forest):

1. Training: Feed historical HR data (1000 employees who left/stayed)
2. Validation: Test on new 200 employees to check accuracy

For AI Models (GPT-4o, Claude):

1. Pre-training: Reads internet to learn language
2. SFT (Fine-Tuning): Taught to answer HR queries like "Write a SQL query for attrition"

3. Two Categories We Used

A. Machine Learning Models (Structured Data)

- Examples: Logistic Regression, Random Forest, XGBoost, LightGBM

- Use in our project: Attrition Prediction, Performance Score Prediction

- Pros: 90% accuracy on HR tables, fast, explainable to HR managers

- Cons: Can't understand text feedback

B. AI General Models (Unstructured Data)

- Examples: GPT-4o, Claude 3.5 Sonnet, Gemini 1.5 Pro

- Use in our project: HR Chatbot, Policy Q&A, Summarizing feedback

- Pros: Understands human language

- Cons: Expensive, needs internet, data privacy risk for client data

4. Final Model Selection for Our Tasks